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Roll No

EX-8001-CBGS

B.E. VIII Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Computer-Aided Design of Electrical Machine

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. With the help of flow chart, explain the synthesis method of machine design problems.
2. Derive how a mathematical formulation can be done for a 3-bus system.
3. Explain Unconstrained optimization problem by using any one method.
4.
 - a) Explain how the variables are selected for optimal design of 3 phase alternator.
 - b) Explain different constraints in synchronous machine design.
5. Why do the turbo alternators have cylindrical rotor instead of salient pole rotor. Determine for a 12.5 MVA, 6.6kV, 2 pole, 50Hz, 3 ϕ alternator suitable values for diameter at air gap, length of core, number of stator slots and conductors. Assume approximate values for missing design constants.

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6. Develop the algorithm for detail design of transformer.
7.
 - a) Explain the method of designing of power transformer for minimum losses and maximum efficiency.
 - b) What are the direct and quadrature axis reactances and how they can be determined.
8. Write short note on any two of the following
 - a) Mathematical programming method
 - b) Objective function and constraint function of 3 ϕ induction motor
 - c) Mathematical formulation of design equation of 3 ϕ alternator.
